

National Land Acquisition & Management System

Smart India Hackathon (SIH) 2026 Technical Report

MEMBER PROFILE: Member 4

Technical Role: GIS Engine, GeoJSON & Mobile Field UI Developer

Primary Focus: GeoJSON coordinate geometry validation checks, React Leaflet map layers rendering & custom styling

1. Functional Domain & Work Done

As part of the NLAMS development team, Member 4 has taken ownership of the functional module targeting GIS Engine, GeoJSON & Mobile Field UI Developer. The following components have been fully implemented and integrated into the live workspace:

- **GPS Triangulator Integration:** Created the mobile field surveyor coordinate calibration widget, simulating satellite triangulation on the mobile web view.
- **Responsive Mobile Reflows:** Restyled FieldSurvey.jsx with flexible flex columns and touch-friendly buttons for easy field operative data entry.
- **Live Leaflet Visualizer:** Connected active parcel polygons centering around corresponding coordinates based on the selected project.

2. Overall Website Layout & Features

The NLAMS portal operates as a single source of truth under a modernised, high-contrast, official government light theme (saffron, navy, and emerald accents). Key features include:

- **Tricolor Home & Directives:** An entry portal showcasing the LARR Act 2013 roadmap and operational pillars.
- **Bilingual Language Switcher:** A global toggle translating all KPI labels, timelines, Gazette notifications, and Treasury advice receipts dynamically into Hindi and English.
- **Executive Dashboard & GIS:** Renders land parcels as Leaflet map overlay polygons, color-coded by acquisition stage, alongside Recharts predictive dashboards.
- **Automated LARR Workflows:** A pipeline allowing Central, State, and District roles to sign off on stages, instantly generating printable official Gazette templates.
- **Web3 Trust & Compensation Portal:** Simulates smart contract escrow releases with Metamask wallet connectivity, a digital deed SHA-256 validation scanner, and a scrolling block ledger.
- **Mobile-Responsive Field Survey Node:** Allows field surveyors to capture coordinate coordinates using GPS triangulation and report soil classification, structures, and site images.
- **SMS Alert Simulator:** Fires dynamic, floating mobile network text notification alerts during proposal submissions, boundary checks, and DBT escrow releases.

3. Sequential Learning Pathway & Project Roadmap

To ensure a smooth transition from prototype to production for the SIH finals, the following learning benchmarks are mapped for this role:

- **Focus Area:** GeoJSON coordinate geometry validation checks
- **Focus Area:** React Leaflet map layers rendering & custom styling polygons
- **Focus Area:** Dynamic color coding by stage status (Green=Acquired, Orange=Pending, Red=Disputed)
- **Focus Area:** HTML5 Navigator Geolocation API integration for GPS coordinate capturing

4. Judge Defense Recommendations

During the pitch and technical Q&A, the judges will test the integrity of each module. Use the following defense arguments to demonstrate technical authority:

- **Defense Tip:** Explain that physical boundaries submitted by field surveyors are cross-checked against registered cadastral database shapes (BhuNaksha) to reject overlaps.